



BRAINWARE UNIVERSITY

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

TITLE OF THE PROJECT

RELAY BASED SOLAR-POWERED STREET LIGHT **WITH EMERGENCY HOME BACKUP**

A PROJECT REPORT

Submitted by

M.VAMSHI KUMAR

(University Roll no – 22010403035, Student code – BWU\DEE\22\038)

Under the Supervision

of

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Department of Electrical Engineering, Brainware University

IN PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

DIPLOMA IN ELECTRICAL ENGINEERING



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DEPARTMENT OF ELECTRICAL ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that the project entitled '**RELAY BASED SOLAR-POWERED STREET LIGHT WITH EMERGENCY HOME BACKUP**' submitted to the Brainware University for the award of the "**DIPLOMA IN ELECTRICAL ENGINEERING**" is a Bonafide work carried out by "**M.VAMSHI KUMAR**" under supervision of "**Mrs. ANINDITA DAS MONDAL, ASSISTANT PROFESSOR, DEPARTMENT OF ELECTRICAL ENGINEERING**". The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Date: / / 2025.

ACKNOWLEDGEMENT

We would like to extend our sincere appreciation for the insightful ideas and contributions regarding solar street lights. The knowledge shared has provided us with a deeper understanding of how to effectively harness renewable power sources, enabling us to make more sustainable choices for the future.

We now see the immense benefits of advancing street lighting systems, not only in terms of energy efficiency but also in reducing environmental impact. The adoption of such innovative technologies will pave the way for more cost-effective, eco-friendly, and reliable lighting solutions in urban and rural areas alike.

Your input has been instrumental in helping us explore more effective ways to improve system performance, and we are confident that these improvements will enhance the overall quality of the surrounding areas. We greatly value this shared knowledge, which will guide us in implementing smarter and greener solutions for our communities.

Once again, thank you for your valuable support and for helping us move closer to a brighter, more sustainable future.

ABSTRACT

The project is about creating an advanced solar street light system that also acts as a backup power source during household power cuts. In the daytime, the solar panels charge a battery, and at night, that stored energy powers the street light. What made the system so special, is that the system didn't use any sensors like LDRs. The system uses a relay-based mechanism that automatically controls the light depending on sun or whether it's day or night. This keeps the system simple, reliable, and fully automatic.

One of the key features that been added is a 200W inverter. During power outage, the inverter allows the stored solar energy to be used in nearby homes for essential needs like lighting. And in case the battery gets too low, the system can easily switch to AC power (but not automatic, have to switch over manually), so the street light continues to function without any interruption.

The project is deemed to maintain to be both eco-friendly and useful during emergencies. It helps to save electricity by using solar power, and it also gives support when there's a power cut at home. The system has been tested, and everything worked well. Even though it's just a prototype, it shows a better and smart way to light streets and help homes at the same time. The idea can be helpful in many places, especially where people can't afford inverters and areas power cuts are more.

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CHAPTER: I

INTRODUCTION

1.1. BACKGROUND

In recent years, the demand for sustainable and reliable urban infrastructure has grown significantly. Among the various innovations aimed at addressing this demand, solar street lighting systems have emerged as a promising solution. These systems not only provide efficient lighting but also contribute to energy conservation and environmental sustainability. However, the potential of solar street lights extends beyond mere illumination. This study explores a Solar-Powered Street Light with Emergency Home Power Backup system that integrates automated lighting control with emergency power supply capabilities, offering a multifaceted approach to urban safety and energy management. The primary objective of this research is to design and implement a solar street light system that operates autonomously, turning on and off in response to ambient daylight levels. During daylight hours, solar panels harness solar energy to charge the system's batteries. A key feature of this system is its 200W inverter, which enables it to provide electricity to nearby homes during power outages, thereby serving as an emergency lighting solution. This system is designed to seamlessly switch between solar power and AC power based on the availability of electricity in the home, ensuring continuous operation. In scenarios where battery levels are critically low, the street light can directly operate on AC power, maintaining its functionality. This dual-mode operation not only enhances urban safety by ensuring consistent street lighting but also contributes to energy efficiency and sustainability. Furthermore, the system's ability to supply emergency power to homes during outages strengthens community resilience against electrical disruptions. This multifunctional approach addresses both the need for reliable street lighting and the demand for emergency power solutions, making it a valuable addition to urban infrastructure. This introduction sets the stage for a comprehensive exploration of the technical specifications, design considerations, and operational mechanisms of the solar street light system. It also outlines the scope of the study, which includes evaluating the system's performance through rigorous testing and analysis, highlighting its benefits, challenges, and potential for future improvements.

In many developing and rural areas, conventional street lighting systems depend heavily on grid electricity and sensor-based technologies like Light Dependent Resistors (LDRs). These systems can be unreliable, difficult to maintain, and prone to failure in harsh weather or dusty environments. Additionally, grid-based lighting systems contribute to rising energy demands and carbon emissions.

To overcome these challenges, the project introduces a relay-based solar street lighting system that operates without LDRs or microcontrollers, making it simpler, more robust, and cost-effective. The system harnesses solar energy through a 12V solar panel, stores it in a bank of 3.7V batteries, and uses a pair of electromechanical relays to control the switching mechanism. One relay is used to detect day or night conditions based on the voltage from the solar panel, while the other manages the inverter control for potential AC applications.

By eliminating dependency on sensitive sensors and complicated circuits, the system becomes more durable and suitable for areas with minimal technical support. A manual override button is also included to ensure functionality under special circumstances or testing conditions. With an efficient 25–30W LED light, this system ensures reliable illumination during night hours while minimizing energy consumption.

This innovative, sensor-free approach not only enhances system reliability but also promotes sustainable development through renewable energy, making it ideal for rural electrification and off-grid lighting solutions.

1.2. MOTIVATION

The increasing demand for sustainable and cost-effective lighting solutions in rural and underdeveloped areas inspired the development of this project. Traditional solar street lights often rely on sensors like Light Dependent Resistors (LDRs), which can degrade over time due to environmental exposure, require complex electronics for control and many other light sensing errors. These factors not only increase maintenance costs but also limit the reliability of such systems in remote locations.

The project was motivated by the idea of creating a simpler, durable, and low-maintenance solar lighting system that could work efficiently without any sensors or microcontrollers. By

using relay-based switching, the design ensures long-term operation with minimal failure points. The use of voltage-based relay logic allows the system to detect day and night conditions using the solar panel's natural voltage variation, removing the need for light sensors altogether.

Furthermore, the goal was to empower communities with a cost-efficient and replicable solution that could be built, understood, and maintained with basic electrical knowledge. The inclusion of a manual override switch adds flexibility for maintenance, emergency use, or testing.

This project represents a practical step toward decentralized, renewable-powered infrastructure, and encourages innovation in sensor-less automation, especially in regions where reliability and simplicity are critical.

1.3. OBJECTIVE

The primary objective of this project is to develop a smart and energy-efficient street lighting system that enhances public safety while reducing electricity consumption. The system aims to incorporate relay-based automation to control brightness levels dynamically.

By integrating solar panels and other renewable energy sources, the project seeks to reduce dependency on conventional electricity grids, promoting sustainability. Another key objective is to create a cost-effective and scalable model that can be implemented in both urban and rural settings, addressing the diverse lighting needs of different regions.

Additionally, the project focuses on minimizing maintenance costs by designing a self-regulating system capable of detecting faults and optimizing energy use. The system is also designed to enhance security in public areas by ensuring well-lit pathways, roads, and open spaces at night.

Overall, the project aims to revolutionize street lighting infrastructure by making it more efficient and environmentally responsible, contributing to the development of smart and sustainable cities worldwide.

1.4. LITERATURE REVIEW

Solar-powered Street lighting systems have become increasingly significant in the global push for clean, renewable energy. While most existing systems focus on advanced control through microcontrollers and sensor networks, such approaches may be costly and less practical in remote or rural environments. Your project introduces an alternative: a solar street lighting and inverter system operated entirely via electromechanical relays—without the use of Light Dependent Resistors (LDRs), microcontrollers, or automation sensors. This approach fills a practical gap in existing literature and addresses real-world constraints such as cost, maintenance complexity, and component availability.

The majority of existing literature focuses on smart lighting systems utilizing sensors and embedded systems. In the work by [7] Abubakar Mukhtar and Rakesh Kumar (2020), titled “Advance Solar Power LED Street Lighting with Auto Intensity Control,” the authors implemented a solar-powered LED street light controlled by an LDR and a microcontroller from the 8051 family. The design incorporated PWM-based dimming and real-time clock (RTC) synchronization to optimize light intensity during nighttime. While effective in energy conservation, this method depends on precise sensor calibration and software reliability—challenges that can hinder rural deployment.

Similarly, in the paper [5] “A Smart Street Lighting System Using Solar Energy” by Fares S. El-Faouri et al. (2016), the authors introduced a streetlight system powered by photovoltaic panels, with added features such as motion sensing, automatic dust cleaning, and data monitoring via MATLAB/Simulink. Though technologically advanced, this system’s reliance on sensors and algorithmic control demands higher maintenance and technical literacy, making it less suitable for regions with limited infrastructure.

[8] Muhammad Asif Rabbani (2020), in his article “Solar Power Systems and DC to AC Inverters,” provides a comprehensive review of solar inverter types and control strategies. His analysis covers half-bridge, full-bridge, and three-phase inverters, with an emphasis on pulse width modulation (PWM) control. Rabbani highlights the importance of inverters in integrating solar-generated DC with AC systems. Your project simplifies this paradigm by controlling inverter operation through a mechanical relay, reducing reliance on complex switching strategies and enabling more accessible, fault-tolerant design.

Similarly, the research by [6] S. Ould Amrouche et al. (2000), “DC/AC Solar Inverter for Solar Applications,” discusses the design of a 2 kVA sine wave inverter using stepped waveform generation and LC filtering. Their work emphasizes ruggedness and local component availability, aligning with your project’s approach to building reliable, serviceable systems using fundamental electronic components.

In the study [3] “Research and Development of Renewable Energy: Prototype of LED Street Lighting from Solar Energy” by Somsanguan Passago et al. (2020), the team evaluated a solar-powered LED streetlight prototype at Rajabhat Mahasarakham University, Thailand. Using 30W LEDs, 80W solar panels, and 12V batteries, the authors found that solar energy collection consistently exceeded energy usage. Their findings support the feasibility of standalone LED systems, validating the design direction of your project’s LED lighting powered via a battery bank charged from solar energy.

M.R. Vervaart and F.D.J. Nieuwenhout, in their technical manual “Solar Home Systems: Manual for the Design and Modification of Solar Home System Components” (2000), emphasize the importance of simplicity, reliability, and cost-efficiency in off-grid systems. The authors argue that components such as battery regulators and inverters must be easy to repair and replace, particularly in decentralized environments. Your project directly follows these principles by using relays instead of programmable or sensor-based circuits, reducing potential points of failure.

[10] Raveendhra Dogga and M.K. Pathak (2019), in their work “Recent Trends in Solar PV Inverter Topologies,” provide a detailed review of inverter classifications and features such as intrinsic boost capability, high efficiency, and transformerless designs. While these advancements improve power quality and integration with the grid, they also increase system complexity. Your relay-based control design avoids such complications by offering an approachable alternative for low-power, standalone systems.

Comparative Summary:

Aspect	Sensor-Based Systems	Your Relay-Based System
Automation	High (LDR, PIR, MCU)	Manual/Relay Logic
Complexity	High	Low
Maintenance	Requires expertise	Simple relay replacement
Cost	Moderate to high	Low
Ideal Use	Urban smart grids	Rural/off-grid settings

Table - 1.1. Comparative Summary.

1.5. RESEARCH GAP

While solar-powered street lighting systems have been widely studied and implemented, most existing models heavily depend on sensor-based automation—especially Light Dependent Resistors (LDRs) or microcontroller-controlled systems (like Arduino or PIC). These components, although efficient under controlled conditions, face significant limitations in real-world environments:

- **Sensor Degradation:** LDRs and other sensors are prone to failure due to dust, moisture, and long-term exposure to sunlight and weather.
- **Complexity and Cost:** Systems using microcontrollers require coding, debugging, and additional components, increasing cost and complexity.
- **Maintenance Issues:** Sensor- or code-based failures often go unnoticed until the entire system stops working, making maintenance more difficult in rural or low-resource settings.

Despite numerous studies on solar lighting systems, very few have explored simple, reliable alternatives that eliminate sensor-based logic entirely, especially using purely hardware-based control methods like relays. This gap presents an opportunity for innovation.

1.5.1. INNOVATION

The project introduces a novel, sensor-free solar street lighting system using a dual-relay mechanism for switching between charging and lighting modes. Key innovations include:

- **No Sensors or Microcontrollers:** The system uses voltage logic from the solar panel to trigger relay switching, removing the need for fragile or expensive sensors.
- **Relay-Based Automation:** Instead of relying on software, the project uses a simple yet effective hardware-based approach for day-night detection and inverter control.
- **Manual Override:** A built-in switch allows users to manually control the system in case of emergencies, testing, or maintenance—adding flexibility.
- **Rugged and Repairable Design:** The entire system is made with off-the-shelf electrical components, making it easy to build, repair, and maintain even in remote areas.

The innovation bridges the gap between high-tech sensor systems and practical, rural-ready solutions, offering a low-cost, durable, and scalable lighting system that supports sustainable development in off-grid regions.

1.6. METHODOLOGY

Creating a solar-powered street light system has several requirements and steps to follow to be successful. The following are the basic components that are solar panels, batteries, relay module, and inverter circuit. The solar panel is the main source of renewable energy, whereas the battery is used to store energy for night usage. The relays control's the lighting system of day and night light or on and off according to the sunlight availability. The process is initiated with the design of the circuit connecting the solar panels, batteries, inverter circuit, relay, and the light source. The system is totally controllable by the relays; it controls the inverter system and also street light system too. The system follows the protocol to change to AC power when the battery levels are very low for that it needs a human interruption in the street light system for AC operational process.

The completed circuit is tested at various light conditions to approve its performance, which is reliable. In case there is a need for sensor calibration and threshold settings, we can implement or install in this system, the system is capable of handling all the sensors. Thus, the solar-powered streetlight system is not only energy-efficient, user-friendly, but also capable of reducing power wastage to a large extent, effectively covering urban and rural lighting.

1.7. PROBLEM FORMULATION

In many rural and semi-urban regions, access to reliable and sustainable street lighting remains a challenge. While solar street lighting systems offer an eco-friendly alternative to grid-powered lights, most available designs depend on sensors like Light Dependent Resistors (LDRs) or microcontrollers for automatic operation. These components:

- Are prone to environmental damage (dust, moisture, heat),
- Require technical knowledge for setup and maintenance,
- Increase the cost and complexity of the overall system.

Such limitations make it difficult to implement and sustain these systems in resource-constrained or remote locations, where simplicity, durability, and ease of repair are critical.

The formulated problem is:

“ How can a solar-powered street lighting system be designed using only basic electrical components like relays, to eliminate the need for sensors and microcontrollers, while ensuring reliable automatic switching between charging and lighting modes, low cost, easy maintenance, and high durability? ”

This problem seeks a hardware-based solution that is:

- Affordable
- Sensor-free
- Energy-efficient
- Rugged for long-term outdoor use
- Simple enough for local maintenance without advanced technical skills

1.8. REPORT OUTLINE:-

- **CHAPTER 2** is about the **System Description** where theory, block diagram and components are explained.
- **CHAPTER 3** is about **System Implementation** where system hardware schematic and operations are explained.
- **CHAPTER 4** is about the **Results, Analysis & Discussion**, like outcomes and performance review and discussions about it.
- **CHAPTER 5** is about the **Conclusion, Future Scopes, Limitations, References** the last part of the project.

CHAPTER: II

SYSTEM DESCRIPTION

2.1 THEORY:

This project is a solar-powered street lighting system that operates entirely using relay-based logic, eliminating the need for light sensors (LDRs), microcontrollers, or programmable control units. The system is designed to provide automatic switching between day and night modes through simple electromechanical relays, making it robust, cost-effective, and easy to maintain—especially in rural or remote areas.

The setup includes a 12V solar panel connected to a battery pack made up of six 3.7V 4A lithium-ion batteries. During daylight hours, solar energy charges the battery bank. A relay is used to detect the presence of solar input; when sunlight is available, the relay disconnects the LED streetlight, keeping it off to save energy. At night, when the solar panel stops generating voltage, the relay switches and automatically powers the LED streetlight using the stored battery energy.

An additional relay is incorporated to control a DC-to-AC inverter, allowing the system to also provide AC output when needed. A manual switch or button is integrated to override the relay function when manual control is desired. This relay-only approach reduces complexity, avoids sensor failures, and offers a practical solution for developing regions where simplicity, durability, and low maintenance are key factors.

FUNDAMENTAL BLOCK DIAGRAM

2.2. RELAY BASED SOLAR-POWERED STREET LIGHT WITH EMERGENCY HOME BACKUP BLOCK DIAGRAM:

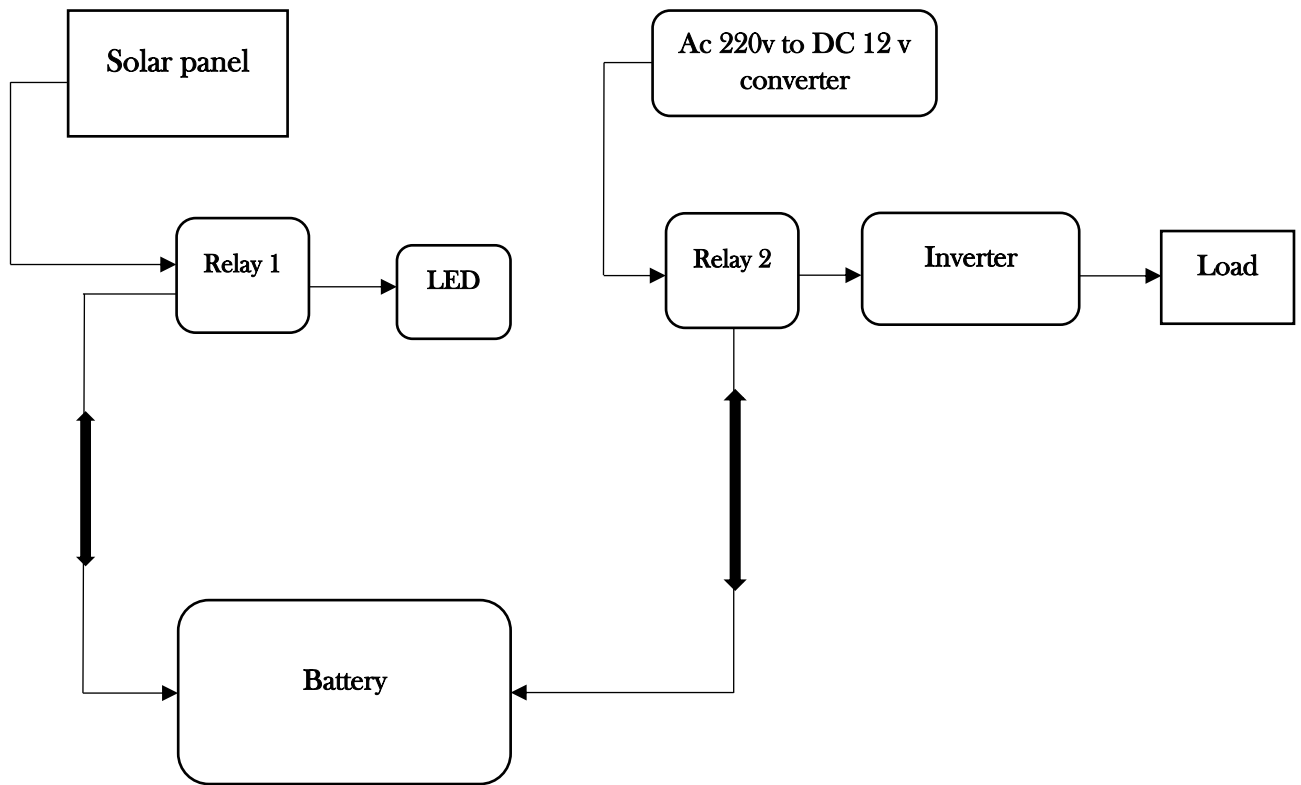


Fig - 2.1 : Block Diagram

2.3. Required Material, Specifications :

➤ Solar Panel:

Solar panels are the core component of our project, serving as the primary source of renewable energy. They work by converting sunlight into electricity through the use of photovoltaic (PV) cells. These cells are typically made from semiconductor materials such as silicon, which generate electrons when exposed to sunlight. As light strikes the cells, the energy excites the electrons, causing them to flow through a circuit and produce direct current (DC) electricity.

This DC electricity can either be used to power various devices directly or be stored in batteries for later use, ensuring a continuous supply of energy even during non-sunny periods. Solar panels are vital for sustainable energy solutions, offering an eco-friendly alternative to traditional power sources. By harnessing the power of the sun, this technology enhances the efficiency, reliability, and overall performance of our system, contributing significantly to a cleaner and greener future.

Sl.no.	Property	Rating
1.	Nominal peak voltage (W)	5w (+-)5%
2.	Peak power voltage (Vmp)	10.8V
3.	Peak power current (Imp)	0.62A

Table - 2.1 : Description Of Solar Panel



Fig - 2.2 : Solar panel

➤ **Transformer (12-0-12, 3A):**

The transformer plays a crucial role in stepping down the high voltage (220V AC) from the mains supply to a safer, lower voltage (12V AC) suitable for powering our circuit. It is specifically designed with a centre-tapped secondary winding, which provides two 12V outputs, along with a common ground. This configuration is essential for creating a dual power supply, allowing for both positive and negative voltage rails, which are often needed in various electronic circuits for stable and efficient operation.

The transformer is rated at 3A, which indicates its maximum current handling capacity, ensuring that it can safely supply enough power for our system's needs. By converting high voltage AC to low voltage AC, the transformer enables the efficient and safe functioning of other components in the circuit, contributing to the overall reliability and performance of the system.

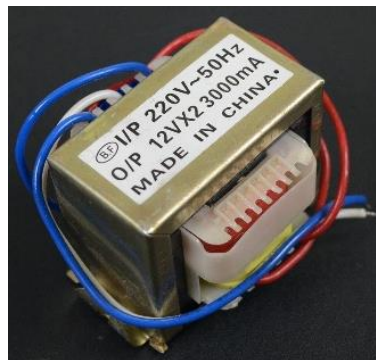


Fig: 2.3. Transformer (12-0-12, 3A)

➤ **Voltage booster XL6009E1 (3v to 35v DC 2-3A):**

The XL6009E1 voltage booster is a vital component in our system, responsible for stepping up the voltage from a lower DC input to a higher DC output. With an adjustable output range from 3V to 35V and a current capability of 3-4A, it ensures the system operates efficiently, even when the input voltage from the solar panels or batteries is low.

This device is designed to boost the voltage to levels required by various components within the system, compensating for fluctuations in input voltage and maintaining stable performance. Whether the energy input is fluctuating due to changes in sunlight or battery

levels, the XL6009E1 ensures that the system receives a consistent, higher voltage for reliable operation. This flexibility is crucial for optimizing energy use and ensuring the longevity and effectiveness of the entire system.



Fig - 2.4 : Voltage booster XL6009E1 (3v to 35v dc 3-4A)

➤ **Inverter circuit (200 Watt):**

The inverter circuit is an essential component that converts the DC electricity stored in the batteries into AC electricity, which is required for powering standard AC devices. With a capacity of 200 watts, it is designed to efficiently handle the energy demands of small household appliances and emergency lighting systems.

This inverter ensures that DC power, typically generated and stored by solar panels or batteries, is transformed into a usable AC form, making it compatible with most household devices. The 200-watt capacity allows for consistent performance in low-power applications, providing a reliable power source during power outages or in off-grid scenarios. By efficiently converting DC to AC, the inverter helps to extend the range and versatility of the system, making it suitable for a variety of practical uses.



Fig - 2.5 : Inverter circuit (200 watt)

➤ **Lithium batteries (3.7v):**

Lithium batteries (3.7V) are crucial components in the solar power system, as they store the energy generated by the solar panels during daylight hours for later use. These rechargeable batteries offer a stable and reliable power supply, ensuring that the system continues to operate even when the sun is not shining, such as at night or during cloudy weather.



Fig - 2.6 : Lithium batteries (3.7v)

Lithium batteries are preferred in this context due to their high energy density, meaning they can store more energy in a smaller space compared to other battery types. This results in a more compact and efficient system. Additionally, lithium batteries have a long lifespan and require low maintenance, making them a cost-effective and durable option for energy storage. Their ability to endure numerous charge and discharge cycles without significant degradation ensures that the system can operate reliably for extended periods, reducing the need for frequent replacements.

➤ **Diode (IN4007), Resistors (1 ohm, 10 ohm):**

The IN4007 diode and resistors (1 ohm, 10 ohm) are crucial components in ensuring the stability and protection of the circuit in a solar-powered system. The IN4007 diode allows current to flow in only one direction, preventing backflow of current that could damage sensitive components such as the battery or solar panel. This reverse current protection is especially important during periods when the solar panel isn't generating power, like at night or on cloudy days, as it prevents the battery from discharging itself.

On the other hand, the resistors (1 ohm, 10 ohm) are used to control the current flow within the circuit, ensuring components operate safely within their limits. The 1-ohm resistor provides minimal resistance, allowing for slight current regulation, while the 10-ohm resistor offers more resistance, helping to significantly reduce the current flow. This helps prevent components from being overloaded, ensuring their longevity and proper functioning. Together, these components work to maintain a stable and efficient power flow, protecting the system from potential damage and ensuring that it operates reliably over time

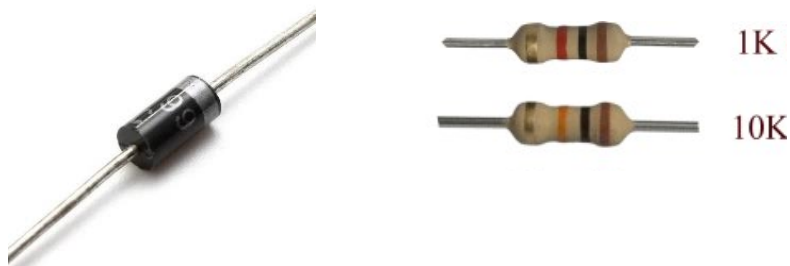


Fig - 2.7 : Diode (IN4007), Resistors (1 ohm, 10 ohm)

➤ **Capacitor (25v 1000uf):**

The capacitor (25V 1000 μ F) plays a vital role in maintaining the stability of the circuit by storing and releasing electrical energy. Capacitors help smooth out voltage fluctuations that can occur due to varying power inputs or load changes within the system. By temporarily storing energy when the voltage is higher than needed and releasing it when the voltage drops, the capacitor helps maintain a more consistent voltage supply. This process significantly improves the overall reliability of the system.

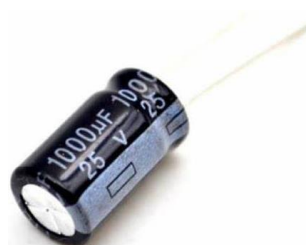


Fig - 2.8 : Capacitor (25v 1000uf)

In solar-powered systems, where energy generation can be intermittent, the capacitor ensures that the connected components, such as batteries or other electronics, receive a stable

voltage, preventing sudden drops or spikes that could cause malfunctions or damage. The 25V 1000 μ F capacitor is well-suited for this purpose, with its high capacitance value allowing it to store a large amount of energy and efficiently smooth out voltage fluctuations.

➤ **Relay (12V):**

The 12V relay is an electrically operated switch that plays a crucial role in managing power sources within a system. It allows the circuit to switch between different power sources, such as solar power and AC power, based on the availability of energy. When the solar power is insufficient, the relay can automatically switch to AC power, ensuring that the system continues to operate without interruption.

This capability helps maintain a seamless operation and uninterrupted power supply. The relay ensures that the system efficiently transitions between power sources, allowing for continuous energy flow without manual intervention. This is especially useful in solar-powered systems, where energy availability can vary throughout the day, ensuring the system remains powered even during cloudy weather or at night.

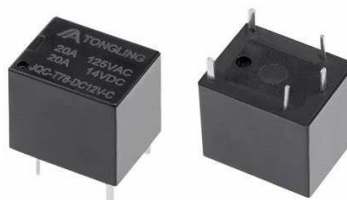


Fig - 2.9 : Relay (12v)

➤ **Switch (on-off):**

The on-off switch provides a simple yet essential function in controlling the power flow within the circuit. This manual switch allows users to easily turn the system on or off as needed, giving them direct control over the operation of the circuit.

When the switch is in the off position, it disconnects the power, preventing any current from flowing through the system. When turned on, it allows the circuit to function normally, enabling the flow of electricity to the connected components.



Fig - 2.10 : Switch (on- off)

This manual control is important for managing energy use, troubleshooting, or performing maintenance on the system.

➤ **220v AC to 12v DC module:**

The **220V AC to 12V DC module** is an essential component that converts high-voltage alternating current (AC) from the mains power supply into a lower, safer direct current (DC) voltage of 12V. This conversion is necessary for charging the batteries in the system, as they typically require DC power to charge.

The module ensures that even when solar power is unavailable, such as during the night or on cloudy days, the system can still charge the batteries using AC power from the grid. This provides a reliable backup charging source, ensuring the batteries remain charged and the system continues to function smoothly regardless of solar energy availability.



Fig - 2.11 : 220v AC to 12v DC module.

➤ **Light (12v):**

The **220V AC to 12V DC module** is an essential component that converts high-voltage alternating current (AC) from the mains power supply into a lower, safer direct current (DC) voltage of 12V. This conversion is necessary for charging the batteries in the system, as they typically require DC power to charge. The module ensures that even when solar power is

unavailable, such as during the night or on cloudy days, the system can still charge the batteries using AC power from the grid. This provides a reliable backup charging source, ensuring the batteries remain charged and the system continues to function smoothly regardless of solar energy availability.



Fig - 2.12 : Light (12v).

➤ **PVC Pipes:**

The PVC pipes are essential for the safe and organized routing of electrical wiring within the lighting system. These pipes act as protective tubes, shielding the wires from potential damage caused by environmental factors such as moisture, sunlight, or physical contact. By using PVC pipes, the system can withstand outdoor conditions while ensuring that the wiring remains secure and free from wear and tear. Additionally, these pipes help maintain a clean, neat installation, preventing the wiring from becoming tangled or disordered, which could pose a risk of short circuits or other issues.



Fig - 2.13 : PVC Pipes.

➤ **Bulb Holder:**

The bulb holder is a critical the bulb holder serves as a key component for ensuring the proper placement and functioning of the LED bulbs. Designed to securely hold the bulbs, it provides a stable mount that prevents them from becoming loose or detached over time. The holder also facilitates easier installation and maintenance, as it allows for quick bulb replacement when necessary.

A well-designed bulb holder can also improve the heat dissipation of the LED bulbs, which contributes to their longer lifespan and improved efficiency component that securely holds the LED bulbs in place. It ensures that the bulbs are properly connected and stabilized, preventing them from falling or becoming loose during operation. The holder also allows for easy replacement or maintenance of the bulbs when needed.



Fig - 2.14 : Bulb Holder.

➤ **LED Bulb (9W, 15W):**

The LED bulbs (9W, 15W) are integral to the lighting system, offering exceptional energy efficiency and long operational life. Unlike traditional incandescent or fluorescent bulbs, LEDs consume significantly less power while providing the same level of brightness. This makes them highly cost-effective, especially for applications requiring long hours of continuous lighting, such as streetlights or outdoor lighting systems.

The 9W and 15W options provide flexibility, as the 9W bulb is suited for more ambient or low-light applications, while the 15W bulb delivers higher brightness for larger or more demanding

areas. Both types are built to last, with a lifespan that far exceeds traditional lighting options, resulting in fewer replacements and lower maintenance costs.



Fig - 2.15 : LED Bulb (9w, 15w)

➤ **Colour LED Bulbs:**

The **colour LED bulbs** add a creative and versatile element to the lighting system, allowing users to customize the lighting effects for different occasions or atmospheres. These bulbs come in a variety of colours, such as red, blue, green, and multicolour, enabling the creation of vibrant, dynamic lighting displays. Ideal for decorative purposes, colour LEDs are perfect for outdoor events, festivals, or simply enhancing the ambiance of a space.

They also retain the energy efficiency and long lifespan of standard LED bulbs, ensuring that even with colourful effects, the lighting system remains cost-effective and low-maintenance. The ability to switch between different colours also provides users with a flexible and dynamic lighting solution.



Fig - 2.16 : Colour LED Bulbs

➤ **PCB circuit board:**

The PCB circuit board (Printed Circuit Board) serves as the foundation for assembling and connecting all the electronic components in the system. It provides a stable and organized platform, where the various components, such as resistors, diodes, capacitors, and connectors, are securely mounted and electrically connected.

The PCB ensures that the circuit operates reliably by minimizing the risk of loose connections and short circuits. It also helps manage the flow of electricity through the components, supporting the overall functionality of the system. The PCB's design and layout are crucial for optimizing performance, reducing interference, and ensuring that all components interact seamlessly.

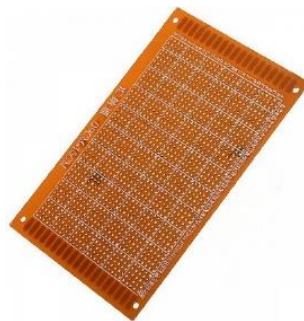


Fig - 2.17 : PCB Circuit Board.

➤ **Wires:**

Wires are essential components in any electrical system, serving as the conductive pathways that connect different components within the circuit. Made from materials such as copper or aluminium, which have high electrical conductivity, wires enable the efficient transmission of electrical signals and power throughout the system. They ensure that electricity flows smoothly between various parts of the circuit, such as from the power source to the batteries, from the relay to the light, or from the solar panel to the charge controller.

The use of high-quality wires helps to minimize voltage drop, reduce energy loss, and improve the overall efficiency and reliability of the system. Wires come in various gauges (thicknesses) depending on the current load they are designed to carry, ensuring that they can safely handle

the electrical demands of each component in the circuit without overheating or causing short circuits.



Fig - 2.18 : Wires.

➤ **12v 25w led strip/sheet light:**

The 12V 25W LED strip/sheet light is designed to provide efficient and bright illumination for outdoor areas such as roads or streets, ensuring clear visibility at night. Powered by the rechargeable battery, this LED light operates during nighttime or when solar power is not available, automatically turning on when the relay switches to load.

The 25W power rating ensures that the light is bright enough to illuminate larger areas, making it ideal for street lighting applications. Using LED technology, the light is highly energy-efficient, consuming less power while delivering consistent brightness, which helps extend the life of the rechargeable battery. This system ensures that the area remains well-lit throughout the night without wasting energy or requiring frequent maintenance.



Fig - 2.19 : 12v 25w LED panel light sheet.

2.4. APPARATUS TABLE WITH QUANTITY WITH HARDWARE CONFIGURATION : -

Here we need:

APPARATUS	QUANTITY
1. Transformer (12-0-12 -3A)	1
2. Solar panel (11v 0.60amp)	1
3. Voltage booster XL6009E1 (3v to 35v dc 3-4A)	1
4. Inverter circuit (200 watt)	1
5. Lithium batteries (3.7v)	6
6. Diode (IN4007),	2
7. Resistors (1k ohm or 10k ohm)	3
8. Capacitor (25v 1000uf)	2
9. Relay (12V)	2
10. Switch (on-off)	4
11. 220v AC to 12v DC charging module.	1
12. Light (12v)	1
13. PCB pipes	2
14. Bulb holder	1
15. LED bulb (9w,15w)	1
16. Colour led bulbs (red, blue, green)	3
17. 3pin socket (6A)	1
18. PCB circuit board	1
19. Wires	As per required

Table – 2.2. : Apparatus/ Materials/ Equipment's with Quantity Table.

2.5. Price of Components :-

APPARATUS	QUANTITY	PRICE
1. Transformer (12-0-12 -3A)	1	210/-
2. Solar panel (11v 0.60amp)	1	300/-
3. Voltage booster (3v to 35v dc 3-4A)	1	50/-
4. Inverter circuit (200 watt)	1	356/-
5. Lithium batteries (3.7v)	6	360/-
6. Diode (IN4007),	2	2/-
7. Resistors (1k ohm or 10k ohm)	3	3/-
8. Capacitor (25v 1000uf)	2	30-20/-
9. Relay (12V)	2	40/-
10. Switch (on-off)	4	40-60/-
11. 220v AC to 12v DC charging module.	1	150/-
12. Light (12v)	1	150/-
13. PCB pipes	2	80+50/-
14. Bulb holder	1	80/-
15. LED bulb (9w,15w)	1	100/-
16. Colour led bulbs (red, blue, green)	3	15/-
17. 3pin socket (6A)	1	80/-
18. PCB circuit board	1	35/-
19. Wires	As per required	Estimation 150/-
20. Box	1	70/-
Total amount -	-	2301/-

Table – 2.3. : APPARATUS/MATERIALS/EQUIPMENTS WITH QUANTITY AND PRICE TABLE.

CHAPTER : III

SYSTEM IMPLEMENTATION

3.1. SCHEMATIC CIRCUIT HARDWARE IMPLEMENTATION & MODIFICATION

Hardware Implementation:

Circuit diagram : -

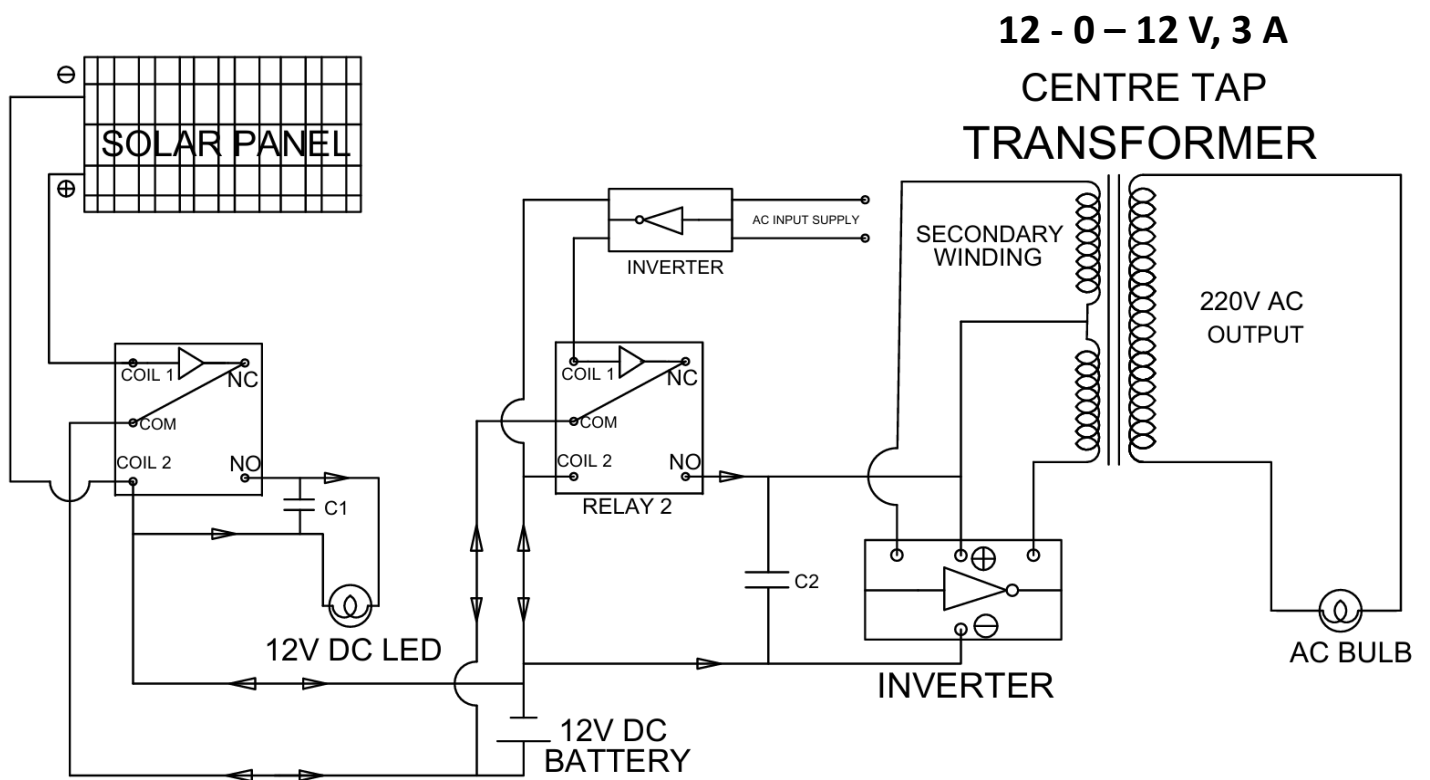


Fig – 3.1. : Schematic Circuit Diagram.

3.2. Operation of the circuit:-

➤ Relay 1:

1. Solar panel will convert light energy into electrical energy, this electrical energy in DC form connect to the coil 1 and coil 2 of 12V relay as shown in figure.
2. Then connect the battery positive terminal to the common point of relay and negative terminal to the coil 2 of relay.
3. Then connect 12V led, positive side of LED to normally open (NO) side of relay and negative side to the coil 2 of relay.
4. Then connect a diode 1N4007 between the two terminals of coil 1 & normally close (NC) of 12V relay as shown in circuit diagram.
5. At last, connect the common point to the normally close (NO) of the relay.

❖ **NOTE: - We have completed the automatic circuit of day and night led on off of street light.**

➤ Relay 2:

6. Now connect the battery positive terminal to common point of relay & negative terminal to coil 2 point of relay.

❖ **NOTE: - We have connected the same battery used for relay 1.**

7. Then we have to connect 220v AC to 12v DC converter to the relay, so the output side of this converter which is 12v dc has a positive and negative side both, so connect positive terminal to the coil 1 point of relay and negative side to the coil 2 point of relay.

8. Now we will be connecting the inverter to the relay 2, take out the two terminals, one from normally open (NO) & another from coil 2.

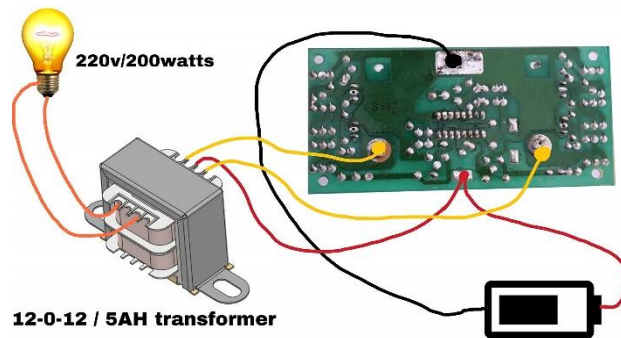


Fig - 3.2. : Circuit Diagram of Transformer & Inverter Circuit.

9. At last, connect the output terminals to the bulb holder and fix a bulb of 9w or 15w LED bulb.

➤ **Operation of the Solar-Powered Street Light with Emergency Home Power Backup with Emergency Power Supply :-**

This system is designed to provide automatic street lighting, ensure energy efficiency, and supply emergency power to nearby homes during power outages, all through intelligent control using Relay 1 and Relay 2.

1. Daytime Operation:

- Solar panels absorb sunlight and convert it into DC power.
- The generated power is used to charge the 12V battery while also activating Relay 1, keeping the LED street light OFF during daylight.
- Meanwhile, the battery stores energy to be used at night or during power failure.
- The system is fully solar-powered during the day, ensuring energy conservation and charging backup storage.

2. Night time Operation:

- When sunlight decreases, the solar panel stops generating power, and Relay 1 deactivates.

- This automatically switches ON the LED street light, powered directly from the charged 12V battery.

- The system ensures uninterrupted street lighting during the night, improving urban safety.

3. Emergency Power Supply (Relay 2 Operation):

- The system includes a 200W inverter connected via Relay 2, which uses the same battery charged during the day.

- Under normal conditions, the inverter remains idle while the home receives power from the main AC supply.

- If a power outage occurs, Relay 2 switches automatically, and the inverter draws power from the battery to supply 12V DC and run essential home appliances or lights.

- This creates an emergency backup system for nearby homes using the streetlight's solar power storage.

4. Dual Power Supply Mode:

- If the battery level is critically low, the system ensures that the street light does not shut off by switching to AC mains power, keeping the streets safely lit.

- This seamless switching between solar power and AC power is fully automated through relays, ensuring continuous operation without manual intervention.

5. Summary of Operation:

- **Day:** Solar charges the battery, street light OFF.

- **Night:** Battery powers street light, ensuring safety.

- **Power failure:** Inverter activates, supplying homes with backup power.

- **Low battery:** System switches to AC to keep the street light ON.

This smart, hybrid system enhances urban resilience, supports sustainability, and offers a reliable, automatic solution for both public lighting and emergency home power.

CHAPTER : IV

RESULTS, ANALYSIS & DISCUSSION

4.1. RESULTS:-

Working time	1 hour
Charging time	2-3 hours by adapter, 3-6 hours by solar as per intensity of sunlight and condition of weather.
Battery capacity	12 V 4000 mah (NOTE: - if we increase the battery capacity working time and charging time will also increase).
Led life	10000 hours
Brightness	2600 lm
Rated power	25W
AC output voltage	199 V
AC output wattage	15 W (we tested till)

Table – 4.1. ANALYSED DATA RESULT TABLE.

4.1.1. Project Performance Results :

The relay-based solar street lighting system was tested under controlled conditions, and the following performance metrics were recorded:

- **Working Time:**

The light operates continuously for 1 hour on a full charge with the current battery configuration.

- **Charging Time:**

Via Adapter: 2–3 hours

Via Solar Panel: 3–6 hours, depending on sunlight intensity and weather conditions.

- **Battery Capacity:**

12V, 4000mAh

(Note: Increasing the battery capacity will proportionally increase both working and charging durations.)

- **LED Life:**

The high-efficiency LED used has an operational life of up to 10,000 hours.

- **Brightness:**

The system delivers a luminous output of 2600 lumens, sufficient for street and outdoor applications.

- **Rated Power of LED Light:**

The LED operates at 25W, aligning with energy-saving standards.

- **Inverter AC Output:**

Voltage: 199V AC

Tested Output Wattage: 15W

(The inverter was successfully tested with a 15W AC load.)

4.2. ANALYSIS & DISCUSSION:-

1. System Functionality Overview:

The developed solar street lighting system utilizes relays for switching between daytime charging mode and nighttime lighting mode without relying on sensors like LDRs. This approach leverages voltage-based switching logic, where relays respond to the presence or absence of solar panel voltage.

➤ Two relays are employed:

- **Relay 1** - automatically controls the LED streetlight based on the solar panel's voltage.
- **Relay 2** - automatically controls the inverter system whenever there are cutoffs in connected home.

This purely electrical control system is intentionally designed to be cost-effective, robust, and free from microcontroller dependencies, which can reduce complexity and maintenance overhead.

2. Performance Analysis:-

I. Energy Flow:

During daylight hours, the solar panel charges the 3.7V lithium-ion batteries. At sunset or when solar voltage drops below relay activation threshold, Relay 1 switches, turning on the LED streetlight.

II. Battery Capacity and Backup:

With approximately 12V 4000mah of storage and a 25W LED load, the system can sustain lighting for nearly 1h - 1h 10min continuously.

Battery discharge patterns and inverter efficiency (typically 80–90%) were factored into this estimate.

III. Efficiency:

By avoiding sensor-based switching and using relay logic, energy consumption from control components is negligible.

- Losses occur primarily during:
 - Battery charging/discharging.
 - DC-to-AC conversion in the inverter.

3. Advantages Observed:-

- **Simplicity:** Relay-based switching provides a low-tech, reliable solution with fewer failure points.
- **Affordability:** Eliminating sensors and controllers reduces cost, making it suitable for rural or low-budget applications.
- **Scalability:** This design can be expanded by adding more battery banks or LED units using similar relay logic.

4. Comparison with Sensor-Based Systems:-

PARAMETER	RELAY - BASED SYSTEM	SENSOR - BASED SYSTEM (e.g., LDR)
Cost	Low	Moderate
Complexity	Simple	Moderate
Accuracy of Switching	Voltage-based (dependable)	Light intensity-based (variable)
Maintenance	Low	Medium

Table – 4.2 : COMPARISON WITH SENSOR-BASED SYSTEMS TABLE.

CHAPTER: V

CONCLUSION, FUTURE SCOPES, LIMITATIONS

5.1. Conclusion: -

This project presents a cost-effective and energy-efficient street lighting system that optimizes power consumption while ensuring proper illumination. By utilizing a relay-based control mechanism instead of sensors, the system reduces complexity and maintenance costs, making it a practical solution for urban and rural environments alike. The implementation of this system can lead to significant energy savings, reduced operational expenses, and improved safety for pedestrians and motorists. Additionally, the integration of solar-powered lighting solutions makes this system a sustainable and eco-friendly alternative to traditional streetlights. With its scalability and adaptability, this project has the potential to contribute to smart city initiatives and sustainable infrastructure development worldwide. Ultimately, this innovation helps create a safer, more efficient, and environmentally responsible street lighting network for modern societies.

5.2. Future Scope: -

1 Smart Technology Integration

Incorporating IoT for real-time monitoring and control, enhancing efficiency and maintenance.

2 Enhanced Energy Storage Solutions

Utilizing advanced battery technologies to improve storage capacity and extend operation during low sunlight.

3 Adaptive Lighting Systems

Developing lights that adjust brightness based on traffic conditions to optimize energy use and safety.

4 Global Expansion

Expanding the deployment of solar street lights in urban and rural areas worldwide, particularly in regions with limited access to electricity.

5 Enhanced Connectivity Features

Incorporating features like Wi-Fi hotspots or emergency communication systems can transform solar street lights into multifunctional urban assets.

5.3. Limitations: -

1 Dependence on Solar Energy:

The system's performance is reliant on solar energy, which can be affected by weather conditions, such as cloudy days or rainy periods, leading to reduced energy production and potentially insufficient battery charging.

2 Battery Capacity and Lifespan:

The effectiveness of the system is dependent on the capacity and longevity of the batteries used. Over time, batteries may degrade, leading to reduced energy storage and shorter operational periods during low sunlight or nighttime.

3 High Initial Installation Costs:

The cost of installing solar panels, batteries, inverters, and the necessary infrastructure can be high, making it challenging for some communities or municipalities with limited budgets to adopt the system.

4 Limited Power Output for Homes:

While the 200W inverter enables emergency power supply to homes during outages, the power provided may be insufficient for powering multiple devices or large households, limiting its effectiveness in emergencies.

5 Maintenance and Monitoring Requirements:

The system requires periodic maintenance, including cleaning the solar panels, checking battery health, and ensuring the inverter and other components are functioning correctly, which could lead to increased operational costs and labour.

6 Reliability of Switching Mechanism:

The transition between solar power and AC power might introduce potential issues if the switching mechanism fails or introduces delays, potentially affecting the reliability of the system during critical moments.

7 Vulnerability to Environmental Conditions:

External factors such as extreme temperatures, dust accumulation on solar panels, or physical damage from storms can degrade the performance of the system, requiring additional preventive measures and repairs.

8 Limited Scalability:

The system might be suitable for small to medium-sized areas but may require further investment and infrastructure for larger-scale applications, making it less feasible for wide deployment in large urban areas without substantial upgrades.

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